1) Define cloud computing and explain any four characteristics that make it different from traditional on premise computing

A. Defination: cloud computing is the delivery of computing services such as servers, storage databases networking and software over the internet instead of maintaing physical infer astructure on premises

Four characteristics:

1) on-demand self service: user can acess computing resource whenever required without directly contecting the service provides

2) Broad network access: cloud services are accessiable throught the internet using laptops

3) Resource pooling: computing resources are shared among multiple users; improving resource

4) Rapid & lasticity: Resources can be increased or decreased quickly according to demand

) Trace the evolution of cloud computing from hardware evolution.

Cloud computing developed through serval technological

1) Hardware evolution: Powerful computers and servers made it possible to process large amount of data efficiently

2) Virtualization: A single physical server can run multiple virtual machines and reducing hardware cost

3) Internet Evolution: Faster and more reliable internet connections enabled users to access computing resources remotely.

4) Web software Evolution: Web applications & web services allowed users to access software through browsers without instally it locely.

3) Server virtualization: It is a technology that divides one physical server into multiple virtual machines each virtual machine run its own operating system and applications.

Role of virtualization in cloud computing:

1) Improving hardware utilization by running mutiple virtual machines on one physical server
2) Reduces hardware and operational cost
3) Supports flexible and scalable cloud infrastructure

Virtualization	Cloud computing
* it is an enabling technology	* It is service delivery model
* creates virtual machines from physical hardware	* Delivers computing resources over a internet
* Mainly improves hardware	* Provides on-demand access to computing services
Eg: VMware	Eg: Amazon web services

A data center is a physical facility containing servers, storage devices, network equipment and power and cooling system.

Roles of data center:-

1) Data storage:- stores large amount of user data & application information.

2) Processing power:- Runs applications, data bases and virtual machines

3) Networking:- connect servers and enables users to access cloud services over the internet

4) Security and backup:- uses security controls, backups and recovery mechanisms to protect data.

5) Availability and Scalability:- ~~Security~~ Controls, ~~backups~~, an Redandant systems help maintain service availability. while additional resources support increasing work loads.

Egs:- Aws data centers provide computing

6) Differeniate Iaas, Paas, Saas & xaas with suitable Example.

Service model	Full Form	Real-world Example
Iaas	Infrastructure as a service	Amazon EC2
Paas	Platform as a Service	Google apps Engines
Saas	Software as a service	Email
Xaas	Anything as a Service	Dropbox Istorage as a service

Explanation:

Iaas: Users manage there operating systems and applications, while the provider manages the underlying infrastructures

Paas: Developers focus on building applications without managing the underlying servers.

Saas: User access complete software applications through a browser or app

Xaas:

A broad term covering services such as storge security, networking and software delivered over the Internet

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